

Regression of Boston House Prices Using Multilayer Perceptron (MLP)

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1 Introduction

This project addresses a regression problem using the Boston Housing dataset. The objective is to predict the **median value of owner-occupied homes (MEDV)** in thousands of dollars using 13 input features.

A Multilayer Perceptron (MLP), a fully connected feedforward neural network, was implemented and evaluated using k-fold cross-validation.

2 Dataset Overview

The dataset contains:

- 506 samples
- 13 input features
- 1 output variable (MEDV)

2.1 Input Features

The 13 features describe characteristics of Boston suburbs such as:

- CRIM – Per capita crime rate
- ZN – Residential land zoning proportion
- INDUS – Non-retail business acres proportion
- NOX – Nitric oxide concentration
- RM – Average number of rooms per dwelling
- TAX – Property tax rate
- PTRATIO – Pupil-teacher ratio
- LSTAT – Percentage of lower status population

2.2 Output Variable

MEDV represents the median value of owner-occupied homes (in \$1000's). Since MEDV is continuous, this is a supervised regression task.

3 Data Preprocessing

3.1 Input-Output Splitting

The dataset was split as:

- X : Shape (506, 13)
- y : Shape (506,)

3.2 Standardization

Standardization was performed using `StandardScaler`. Each fold of cross-validation was standardized separately to prevent data leakage.

Standardization ensures:

- Mean = 0
- Standard deviation = 1
- Faster convergence
- Balanced feature contribution

4 Model 1: Two Hidden Layer MLP

4.1 Architecture

- Input Layer: 13 units
- Hidden Layer 1: 64 units (ReLU)
- Hidden Layer 2: 32 units (ReLU)
- Output Layer: 1 unit (Linear)
- Optimizer: Adam
- Loss Function: Mean Squared Error (MSE)

4.2 Evaluation Method

- 10-Fold Cross Validation
- Shuffle enabled
- Fixed random seed

4.3 Results

- Mean MSE: 11.39
- Standard Deviation: 5.51

The model achieved strong performance compared to the expected benchmark MSE of approximately 20.

5 Model 2: Single Hidden Layer Network

A fully connected network with one hidden layer and ReLU activation was implemented.

5.1 Results for Different Hidden Units

Hidden Units	Mean MSE
8	29.41
16	17.53
32	14.33
64	12.60
128	11.84

Table 1: Single Hidden Layer Performance

5.2 Discussion

- 8 units resulted in underfitting.
- Increasing the number of units consistently reduced MSE.
- Performance improved significantly between 16 and 64 units.

Selecting the appropriate number of hidden units required experimentation. Too few units reduce model capacity, while too many increase computational cost and risk overfitting.

6 Model 3: Arbitrary Depth Network

A generalized function was implemented to construct networks of arbitrary depth.

6.1 Architectures Tested

Architecture	Mean MSE
[64]	12.75
[64, 32]	12.00
[64, 32, 16]	10.72
[128, 64, 32, 16]	9.34

Table 2: Arbitrary Depth Network Results

6.2 Effect of Depth

Increasing network depth consistently reduced MSE. The deepest architecture tested achieved the best performance:

- Best Model: [128, 64, 32, 16]
- Mean MSE: 9.34

This demonstrates that deeper networks can capture more complex relationships in the data and converge to better solutions.

7 Overall Conclusions

1. The MLP successfully modeled the Boston housing regression problem.
2. Standardization significantly improved convergence.
3. Two hidden layers achieved strong performance ($MSE = 11.39$).
4. A single hidden layer required tuning of hidden units.
5. Increasing depth progressively improved predictive accuracy.
6. The best performing model was [128, 64, 32, 16] with $MSE = 9.34$.

The obtained performance is substantially better than the expected benchmark ($MSE \approx 20$), demonstrating effective architecture selection and hyperparameter tuning.

8 Environment

- Python 3.x
- TensorFlow (Keras API)
- scikit-learn
- NumPy
- Pandas